AMOGH THAKUR.

HARDWARE FOR AI

ECE 510 - CMAN L

1)
→

Number of MACS = Number of Inputs.

As this is a fully connected network. each ~~node~~ neuron will be connected ~~to~~ by all the ~~nodes~~ neuron from previous layer.

Layer 1 ⇒ $784 \times 256 = \underline{200704}$. Layer 2 ⇒ $256 \times 128 = \underline{32768}$

Layer 3 ⇒ $128 \times 10 = \underline{1280}$

2)
→ Sum of MACS = MAC(Layer 1) + MAC(Layer 2) + MAC(Layer 3)

$= 200704 + 32768 + 1280$

Sum of MACs. = $\underline{234,752}$

3)
→ Number of weights (No Biases) is same as number of MACs.

Number of weights = $\underline{234,752}$.

4)
→ Weight memory in bytes = $234,752 \times 4$ bytes.

= $\underline{939,008 \text{ bytes}}$.

5)
→ Total Activation memory = Sum of Neuron from L1, L2, L3 and Inputs × 4 bytes.

$= (784 + 256 + 128 + 10) \times 4$ bytes.

= $\underline{4712 \text{ bytes}}$

6)
→ Arithmetic Intensity = $\dfrac{2 \times \text{total MACs}}{\text{weights bytes} + \text{Activation bytes}}$ $= \dfrac{2 \times 234,752}{939,008 + 4712} = 0.4975$